Adult Income Prediction

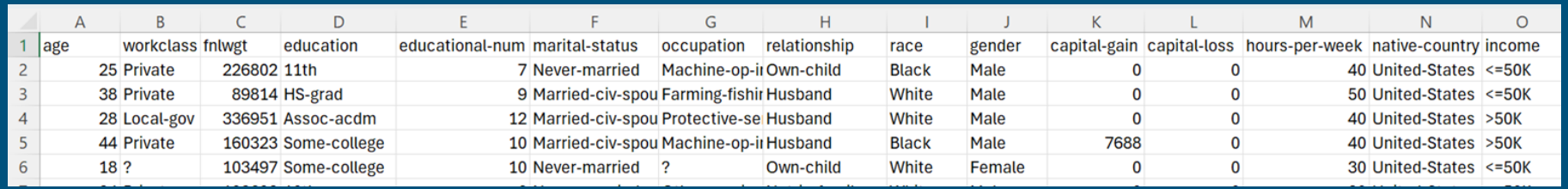
Data Science I | Bonnie Houston

Motivation & Aim

- ➔ Motivation:
  - ◆ Predict whether an adult earns <= 50K or > 50K using demographic features.
  - ◆ Importance: demographic analysis, tax policy, research, credit modeling, and more
- ➔ Aim:
  - ◆ Build a custom decision tree, tailored to the adult income dataset and improve with AdaBoost.
  - ◆ Test whether model is statistically significant using paired t-tests.
  - ◆ Compare against baseline models and determine the best tool for socio-economic prediction and analysis.

Data Source

- ➔ Source: U.S. Census Bureau → publicly available on Kaggle.
  - ◆ 48,843 entries (Adult income dataset)
- ➔ Features:
  - ◆ Continuous: age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week
  - ◆ Categorical: workclass, education, marital-status, occupation, relationship, race, sex, native-country
- ➔ Target: income (<=50K: 76.07%, >50K: 23.93%)
- ➔ Missing values
  - ◆ represented as “?” (7% of entries)



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	age	workclass	fnlwgt	education	educational-num	marital-status	occupation	relationship	race	gender	capital-gain	capital-loss	hours-per-week	native-country	income
2	25	Private	226802	11th	7	Never-married	Machine-op-i	Own-child	Black	Male	0	0	40	United-States	<=50K
3	38	Private	89814	HS-grad	9	Married-civ-spou	Farming-fishi	Husband	White	Male	0	0	50	United-States	<=50K
4	28	Local-gov	336951	Assoc-acdm	12	Married-civ-spou	Protective-se	Husband	White	Male	0	0	40	United-States	>50K
5	44	Private	160323	Some-college	10	Married-civ-spou	Machine-op-i	Husband	Black	Male	7688	0	40	United-States	>50K
6	18	?	103497	Some-college	10	Never-married	?	Own-child	White	Female	0	0	30	United-States	<=50K

Data Preprocessing

- Feature selection:
  - Removed fnlwgt (not predictive) and educational-num (duplicate of education).
  - Filtering improved max F1 score from 51% to 72% for a single decision tree.
- Kept “?” values as a potentially useful predictor.
- Randomly selected 10,000 samples for efficiency and representativeness and shuffled data to eliminate any ordering bias.
- 75/25 train/test split.
- Binning continuous features lowered F1 scores → kept original values.

Methods and Models

- ➔ Data Structure and Approach:
  - ◆ A mix of categorical and continuous features which decision trees handle well.
  - ◆ Trees can model nonlinear decision boundaries and are easily interpretable.
- ➔ Custom Model
  - ◆ Implemented an unpruned decision tree from scratch.
  - ◆ Utilized entropy and information gain.
  - ◆ Max F1 score: 72%

Improvement:

- ➔ AdaBoost
  - ◆ Effective for unbalanced data and reducing weak-learner bias.
    - Used pre-pruning (max_depth = 1) to avoid overfitting.
  - ◆ Optimal performance with ≅ 75 trees (25 also strong and more efficient).
  - ◆ Improved F1 score from 72% to 80%.
- ➔ Feature selection contributed to improvement.

Evaluation Methods

- ➔ Approach:
  - Chose F1 score due to class imbalance.
  - Numeric encoding required for baseline models.
  - Diverse baseline models for fair comparison.
- ➔ Baseline Model Results:
  - Perceptron F1: 66.34% (linear boundary → data is nonlinear)
  - Naive Bayes F1: 75.78% (independence assumption limits performance)
  - Random Forest F1: 79.71% (ensemble → strong predictor)

Statistical Significance

- ➔ Paired t-tests
  - 5 iterations, each using a new random selection of 10,000 samples.
  - same samples and train/test across all models → paired F1 scores.
- ➔ P-values
  - `Perceptron:  0.00134 | Naive Bayes:0.0037 | Random Forest: 0.00055`
    - Changes can cause fluctuations in Random Forest p-value due to similar performance.
- ➔ Conclusion:
  - All p-values < 0.05 → reject all null hypotheses.
  - Model is statistically significant vs. all baseline models.

Project Contributions

- ➔ Produces a reliable prediction of an adult’s income given common demographic features.
- ➔ Helps uncover feature relationships, which are useful across economics, social science, and government policy to help combat socio-economic disadvantages.